

Dhanaraj Nitin Patil

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TECHNICAL SKILLS

Languages: Java, C#, Python, C++, JavaScript, Shell Scripting

Distributed Systems: Apache Kafka, Event-Driven Architectures

Backend & APIs: REST APIs, Spring Boot, Hibernate, Microservices

Cloud & Data: DynamoDB, Redis, OpenSearch, SQL, NoSQL, Postgres, AWS (Lambda, S3)

Frontend: Angular, React, HTML, CSS

Observability & DevOps: New Relic, CI/CD, Jenkins, Git, Rancher

EDUCATION

Master's in Computer Science

San Diego State University, San Diego

Bachelor's in Information & Computer Science

BMS College of Engineering, Bangalore - 2014

WORK EXPERIENCE

Software Engineer III

Lytx, Inc.

June 2023 - PRESENT

(Remote) San Diego, CA

Lytx is a video telematics and fleet management company that helps organizations improve safety, efficiency, and regulatory compliance through data-driven insights.

- Built features and scalability improvements for a Kafka-based device telemetry **ingestion platform processing 1T+ events/day**.
- **Owned fleet APIs backing the Fleet monitoring dashboard**, migrating reads from Redshift to OpenSearch to enable querying up to **3 years of trip data** while **improving data fetch latency by ~50%**.
- **Developed vehicle lifecycle APIs** to create and associate vehicles with drivers and devices across fleet management systems.
- Designed and built an Angular-based release management portal **integrated with backend APIs**, enabling self-service firmware and module uploads, cohort-based targeting, and staged deployments to camera devices.
- **Designed a DynamoDB-OpenSearch write/read architecture** powering real-time device activity dashboards with geofence alerting, operational monitoring, and fast search.
- Modernized legacy SOAP-based interfaces, transitioning to a RESTful HTTP architecture, which improved maintainability, developer adoption, and integration velocity.
- Instrumented systems with **production-grade observability**, monitoring Kafka lag, errors, and query latency using New Relic.
- Leveraged AI-assisted development tools and vibe coding techniques to accelerate development, debugging, AWS/Kafka log analysis, and query optimization while maintaining high production-grade reliability.
- Diagnosed and resolved production issues in large-scale distributed systems through log and metric analysis.
- **Tech: C#, Java, Angular, Multi-Threading, Kafka, REST APIs, Angular, Elasticsearch, AWS, Dynamo, New Relic, Git, Rancher, Argo.**

Software Engineer

Teradata

May 2022 - December 2022

San Diego, CA

- Contributed to a **Configuration Validation tool** that validates and configures systems before executing DM jobs, **reducing job execution time and effort by up to 70%** through pre-checks.
- Researched **JWT-based API authentication** to modernize and potentially replace the existing authentication mechanism.
- Implemented **secure coding best practices**, including input validation, secure session handling, and robust exception management, **improving code quality and developer efficiency by ~60%**.

Senior Software Engineer

Informatica

February 2018 - August 2021

- Upgraded functionality to **extract metadata from millions** of customer records & scan for sensitive data discovery.
- Rolled out a REST API interface with **25+ APIs** for TDM users, which enabled access to most of the features without the UI.
- Designed and scaled microservices powering high-traffic features, improving API latency by ~40%.
- Delivered a **Subject Registry framework**, where multiple data requests such as DSAR, Breach notifications, etc., can be generated over millions of customer data **to comply with data governance rules**.
- Lead a team of 3 engineers for a **seamless upgrade** of the Informatica Data Privacy Management product.
- Designed and authored an algorithm for the highly anticipated **Incremental Scan** feature for both structured and unstructured data sources, which **enhances scan time by 80%** for subsequent scans.
- Created a dashboard with **10 widgets** for subject registry summary to display the data source-wise details of Privacy mandates
- Automated unit and integration tests using JUnit and improved the response time by 45%.

System Engineer

Tata Consultancy Services

June 2014 - February 2018

- Worked on modules in e-Forms manager, which capture and submit insurance applications for a user.
- Introduced a new customer onboarding portal with REST APIs.